THE COLLECTIONS OF FPPF COVERINGS DEFINE SITES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2306, 2310] and proof(s) [[2312, 2324]]; this spec covers the complete environment instead (statement 2306-2310, proof span(s) [(2312, 2324)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original lemma 021C: sources/stacks-chow/tags.txt, source lines 2665–2665. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VH: sources/stacks-chow/tags.txt, source lines 1147–1147. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let Sch_{fppf} be a big fppf site containing S. Then $(Aff/S)_{fppf}$ is a site.*

Proof. Let us show that $(Aff/S)_{fppf}$ is a site. Reasoning as in the proof of Lemma 021C it suffices to show that the collection of standard fppf coverings of affines satisfies properties (1), (2) and (3) of Sites, Definition 00VH. This is clear since for example, given a standard fppf covering $\{T_i \to T\}_{i\in I}$ and for each i we have a standard fppf covering $\{T_{ij} \to T_i\}_{j\in J_i}$, then $\{T_{ij} \to T\}_{i\in I, j\in J_i}$ is a standard fppf covering because $\bigcup_{i\in I} J_i$ is finite and each T_{ij} is affine. □

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